Esher Swanson

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EDUCATION

University of New Hampshire, Durham NH — Anticipated Graduation: **May 2027**

Bachelor of Science in Marine, Estuarine, and Freshwater Biology; Bachelor of Science in Sustainability — GPA: **3.94**

- **Hamel Honors and Scholars College:** Secretary of the Honors Undergraduate Social Committee (HUSC)
- **Dean's List:** *Highest Honors* (Fall 2023, Spring 2024, Spring 2025, Fall 2025) *High Honors* (Fall 2024)

Relevant Coursework: Data Science for Life Sciences (using R) Sustainable Fisheries; Marine Ecology; Field Bioacoustics; Sharks and Bony Fishes; Biostatistics; Principles of Aquaculture; Honors Exploring Sustainability; Critical Thinking; Aquatic Plants in Restoration/Management; Seamanship and Marine Weather; Marine Mammal Biology and Conservation; Data Analysis (using R)

SKILLS

Field Skills: Fish handling and measurement; wading in mudflats, sand, and rocky habitat; cod fishing; hydrophone, speaker, and AudioMoth deployment/recovery; directional microphone recording; quadrat and transect sampling; driving Zodiacs; lobster pot deployment/recovery; temperature, depth, and water quality monitoring; beach seining; electrofishing; sample collection by neuston tow; data recording; VAR2AR receiver prep and deployment; gull nest checks; knowledge of fishing gear types and uses; rainbow smelt sampling and scale collection; eDNA collection; marine weather analysis; ship safety; PIT tag scanning; camera trap deployment, maintenance, and recovery; fish gastric lavage (stomach flushing); bag seining;

Lab Skills: *Artemia* culture (including decapsulated); rotifer culture; knowledge of aquarium systems; water quality analysis; fish husbandry; micropipetting; fish/jellyfish movement (between tanks, etc.); rainbow smelt scale cleaning, mounting, and aging; muscle dissection for stable isotopes; fish stomach content analysis; otolith identification; pellet pressing; sample preparation for bomb calorimetry; dissecting otoliths from fish

Technical Skills: RStudio; Microsoft Office Suite; ImageJ; Raven Pro; Audacity; SeBaView; CaptaVision+

RESEARCH EXPERIENCE

Independent Striped Bass and Atlantic Cod Diet and Prey Energetics Study in Dr. Nathan Furey's Fish and Movement Ecology Lab, Durham, NH — **January 2026 – Present**

- Coauthored our research objectives and proposal to secure funding and necessary study materials
- Caught study species using hook and line fishing with local recreational fishermen
- Investigated stomach content of predatory fish using non-lethal gastric lavage method
- Identified prey fish species by dissecting out their otoliths and identifying those otoliths
- Prepared samples of 10+ species for bomb calorimetry by drying fish, homogenizing samples, and pressing them into pellets
- Studied energetic content of prey using bomb calorimetry technique and compared to literature values
- Performed statistical analysis, interpreted results, and created graphs using Rstudio to communicate results
- Will present results at the 2026 Regional Association for Research in the Gulf of Maine (RARGOM) Conference
- Will present results in 2027 at one or more research conferences including UNH's Undergraduate Research Conference

Independent Aquaculture Study in Dr. Karolina Kwasek's Lab, Durham, NH — **January 2025 – Present**

- Coauthored our study design and proposal to secure funding for our research objectives and necessary study materials
- Investigated the effects of different live feeds on the growth and survivorship of 1200 larval zebrafish to simplify current feeding regimes used both in research labs within many industries and in research labs on my university's campus
- Weighed (both individual and biomass approaches) 420 larval zebrafish and measured lengths using calipers and ImageJ
- Performed statistical analysis, interpreted results, and created graphs using RStudio to communicate results
- Authored a research brief upon personal invitation by Inquiry Journal
- Lead authored a scientific paper to submit to a high impact journal and participate in the publishing process

Field Bioacoustics Class Research at Shoals Marine Lab (SML), Kittery, ME **May 2025 – June 2025**

- Conducted audio playback experiments measuring 21 nesting American herring gull's behavioral responses to sounds in order to both see if they have the ability to recognize their neighboring gull's alarm calls and to see their responses to them
- Recorded and cleaned gull sound files using Raven Pro and Audacity to create playback files to play to nesting gulls
- Created graphs using RStudio and presented these graphs, observed trends, and significant findings to ~80 people at SML

RELEVANT WORK EXPERIENCE

Undergraduate Fisheries Technician (specializing in seabird prey) in Dr. Nathan Furey's Fish and Movement Ecology Lab, Durham, NH ($20 an hour for 3 hours per week) **August 2026 – Present**

- Prepare hundreds of tern prey samples for bomb calorimetry by weighing, measuring, identifying, and drying the fish, homogenizing it into powder, and pressing this powder into a combustible pellet to estimate energy density of prey species

Undergraduate Fisheries Technician in Dr. Nathan Furey's Fish and Movement Ecology Lab, Durham, NH ($15 an hour for 6-10 hours a week) **August 2025 – Present**

- Clean, process, and mount thousands of rainbow smelt (*Osmerus mordax*) scales on microscope slides using pancreatin and a sonic bath, facilitating accurate age determination for future research purposes
- Rainbow smelt sampling fieldwork in Spring 2026 in association with NH Fish and Game: weighed, scanned, measured, sexed, tagged, and collected scales from rainbow smelt at the Squamscott, Oyster, and Winnicut rivers
- Data entry and scale collection organization for all rainbow smelt spring fieldwork
- eDNA collection transect across a salinity gradient in the Piscataqua river and Gulf of Maine
- Identify ages of acoustic telemetry tagged rainbow smelt using cleaned and mounted scale samples

Zoo Crew Volunteer at Omaha's Henry Doorly Zoo and Aquarium, Omaha, NE **January 2018 – August 2023**

- Interacted with and informed thousands of guests on conservation issues and animal facts while answering visitors' questions
- Supervised and led all youth volunteer programs while monitoring success in their goals as part of Zoo Crew Leadership

Aquarium Intern at Omaha's Henry Doorly Zoo and Aquarium, Omaha, NE **May 2022 – March 2023**

- Prepared food for a variety of marine organisms, completed water chemistries, learned plumbing skills and cleaning techniques for filter socks, bio-towers, and tanks made of both glass and acrylic
- Performed routine tasks such as weighing out traces and minors for seawater, scrubbing pipes, and buffering the shark tank
- Sorted and moved jellyfish, learned about their life stages, and aquarium life support systems with bio-towers and sumps
- Data recording during feedings, fed a variety of organisms such as cownose rays, eels, many fish species, and sea turtles

PEER REVIEWED JOURNAL PUBLICATIONS IN PROGRESS

1. **Swanson E**, Robinson J, and Kwasek K**.** Playing with Sea Monkeys: An Examination of the use of Decapsulated Artemia as a Means of Improving Current Zebrafish Culture Practices. **Submitting in 2026**

RESEARCH BRIEF JOURNAL PUBLICATIONS

1. **Esher Swanson** (Mentor: Karolina Kwasek). Playing with Sea Monkeys: An Examination of the use of Decapsulated Artemia as a Means of Improving Current Zebrafish Culture Practices. Inquiry Journal. **Published Dec. 3, 2025**
   - Personal invitation by the journal's senior editor to publish after she found and read my final report for this study

RESEARCH FUNDING AWARDS

Summer Undergraduate Research Fellowship (SURF) for Proposal "Investigating Diets of Striped Bass and Atlantic Cod and the Energy Content of their Prey"; ***Esher Swanson,*** *Nathan Furey ($4000 stipend and $600 expense budget)* **May 2026 – August 2026**

- Awarded by the Hamel Center for Undergraduate Research at UNH thanks to the generosity of the J. Raymond Hepler Endowed Fund established by Mrs. Helen Hepler Clapp and Mr. and Mrs. Bruce G. Brown II

The Elizabeth Gardner Norweb Summer Environmental Studies Scholarship to fund summer research with Dr. Nathan Furey's Fish and Movement Ecology Lab; ***Esher Swanson,*** *Nathan Furey ($3000)* **May 2026 – August 2026**

- Awarded by the Garden Club of America

Undergraduate Research Award (URA) for Proposal "Replacing Rotifers with Decapsulated Artemia as a Means of Improving Current Zebrafish Culture Practices"; ***Esher Swanson,*** *Karolina Kwasek ($1200 stipend and $600 expense budget)* **May 2026 – August 2026**

- Awarded by the Hamel Center for Undergraduate Research at UNH thanks to the generosity of Mr. Dana Hamel

ADDITIONAL WORK EXPERIENCE

AI Trainer for Handshake AI, Remote ($17.00-$19.00 an hour; variable hours up to 80 per week) **May 2026 – August 2026**

- Compare output of AI models to prompt input to evaluate AI competency at performing tasks and fulfilling users' needs

UNH Organic Garden Club Farm Hand, Durham, NH ($12.00 an hour for 20 hours a week) **June 2025 – August 2025**

- Maintain farm spaces, assist in crop production, manage farm stands, and interact with clientele and farm visitors
- Find and complete tasks without oversite, pace outdoor work on peak temperature days, problem solve with limited supplies

Student Staff at Shoals Marine Lab (SML), Kittery, ME ($14.65 an hour + overtime pay for 48+ hours a week) **May 2024 – June 2025**

- Prepared meals for the island population (~90 people) and visiting groups (~45 additional people) with the chefs
- Island maintenance and landscaping including trail cleaning, cleaning and organization of common spaces, dorm turnover, store inventory, and assisting the engineering team with solar field installation
- Viewing guest lectures on marine biology topics, joining dissections, learning from boat captains, and assisting undergraduate researchers with their projects including fish observation, seal surveys, and gull nest checks during any free time I had

Volunteer and Teen Advisory Board for Omaha Public Library, Omaha, NE **May 2019 – August 2023**

- Sorted and shelved books, assisted in running the Lego club and STEM programs at the Bess Johnson Elkhorn Branch
- Advised the library system on library ideas, renovation planning, discord testing, and updates to teen spaces
- Contributed ideas, planning, and execution of teen programming including the idea, execution, and details for the popular and successful Escape from Mr. Lemoncello's Library Escape Room that was promoted by most of the 12 library branches

Office Manager for Earthshade Natural Window Fashions, Omaha, NE **May 2022 – August 2022**

- Received, sorted and opened all mail, processed all checks and customer payments, shipped shade samples out to customers

Office Assistant for Skypark Business Solutions, Omaha, NE ($10 an hour, flexible hours) **May 2020 – August 2021**

- Internship leading to employment; completed a variety of office tasks including mailing dues notices to HOAs, processing dues payments, a variety of organization tasks, processing ACH payments, and filing of paperwork

PRESENTATIONS (presenters listed first and underlined)

What's for dinner? A Comparison of Gulf of Maine Forage Fish Energy Densities, Durham, NH

- **Esher Swanson,** Kimberly A. Lato, Nathan B. Furey; Regional Association for Research in the Gulf of Maine Annual Science Meeting: Navigating Change in the Gulf of Maine - Science, People, and the Next Generation; Poster

November 4, 2026

Playing with Sea Monkeys: An Examination of the Use of Decapsulated Artemia as a Means of Improving Current Zebrafish (Danio rerio) Culture Practices, Durham NH **April 2026**

- **Esher Swanson,** Jessica Robinson, Karolina Kwasek; University of New Hampshire Undergraduate Research Conference: Interdisciplinary Science Symposium; Poster; **Awarded 2nd Place in Biology** **April 22, 2026**
- **Esher Swanson,** Jessica Robinson, Karolina Kwasek; University of New Hampshire Undergraduate Research Conference: All Colleges Undergraduate Research Symposium Sessions 1 and 2; Poster **April 24, 2026**
- **Esher Swanson,** Jessica Robinson, Karolina Kwasek; University of New Hampshire Undergraduate Research Conference: College of Life Sciences and Agriculture Undergraduate Research Symposium; Poster **April 25, 2026**

Marine Debris Across Four Beaches on the Gulf of Maine, Durham, NH **April 23rd, 2025**

- **Esher Swanson,** Anna Hogan, Emily Palermo; University of New Hampshire Undergraduate Research Conference: Sustainability Symposium; Poster; **Awarded 1st Place in our category of the sustainability section of the conference**

HONORS & AWARDS

- University of New Hampshire Honors Program (2023-2027)

- Mary Kaye Eggers Memorial Scholarship (Awarded April 26, 2023; 2023-2027)

- UNH Trustee's Scholarship (Jan 10, 2023-2027)

-Plant Science Ford Scholarship (2026)

-Mavis H. Foster Scholarship (2026

-University of New Hampshire's Hamel Honors and Scholars College Class of 1966 Scholar (2026)

- Nominated to represent the University of New Hampshire for the Udall Scholarship competition (2026)

- Nominated to represent the University of New Hampshire for the Truman Scholarship competition (2026)

- Service Credit Union Impact Foundation Scholarship (2025; applied for and was awarded a second time in 2026)

- Rutman Scholarship (2024; applied for and was awarded a second time in 2025) through Shoals Marine Laboratory

- COLSA Scholarship (2024) through University of New Hampshire

- 3rd place in UNH fencing tournament (2024)

- National TEAM+S TSA Competitor in Kentucky (2023)

- Graduated top 3 in high school class (2023)

- AP Scholar with Distinction (2023)

- Omaha Elks Lodge Most Valuable Student Scholarship (May 15, 2023)

- Elkhorn High School Recognition of Achievement in Outstanding Academic Tutoring (2023)

- Elkhorn High School Recognition of Achievement in Science (2023)

- President's Award for Educational Excellence (2023)

- 2023 EPSF Achieve Scholarship (April 26, 2023)

- Legacy Eyecare Scholarship (April 26, 2023)

- 1st Place at EMC Conference Quiz Bowl Championship (2023)

- Omaha World Herald All Academic Program Honorable Mention (2023)

- Senior Class Board (2022-2023)

- NSAA Academic All-State Award (Fall 2021, Spring 2022, Fall 2022, Spring 2023)

- 4.0 unweighted GPA on 4.0 scale for all 4 years of High School (2019-2023)

- Academic Letter received for all 4 years of High School (2019-2023)

- UNL Regents Scholar (did not accept; Oct 5, 2022)

- UNL CASNR Scholarship (did not accept; Nov 30, 2022)

- UNE Presidential Merit Award (did not accept; Dec 8, 2022)

- UNO Regents Scholar (did not accept; Nov 8, 2022)

- Greater Omaha Area Youth Salute Top Youth Leader from Elkhorn High School (2022)

- Rotary Outstanding Sophomore Award (2021)

TRAININGS & WORKSHOPS

Honors Cocurricular Aquaculture Experience with Dr. Thomas Safford and Dr. Analena Bruce of UNH **November 2025-May 2026**

- will visit aquaculture industry in Maine and the UNH Aquafort
- learn about aquaculture, its stakeholders, and how it is conducted through talking to many industry experts and stakeholders

CITI Program Ethical Handling of Zebrafish in Research Settings **May 5, 2025; Expires May 5, 2030**

CITI Program Research Security Training **May 5, 2025; Expires May 5, 2030**

CITI Program Vertebrate Animal Care and Use for Student Researchers **May 5, 2025; Expires May 5, 2030**

CITI Program Biomedical Research Ethics and Responsible Conduct of Research **May 5, 2025; Expires N/A**

CITI Conflicts of Interest for Investigators on Externally Funded Research **May 14, 2025; Expires May 14, 2029**

CITI Program Ethical Handling of Fish in Research Settings **May 14, 2025; Expires May 14, 2030**

CITI Program IACUC Community Member Course **May 14, 2025; Expires N/A**

CLUBS & INVOLVEMENT

- Intramural Indoor Soccer at University of New Hampshire (2026-present)

- Organic Garden Club at University of New Hampshire (Farm Hand and Executive Board Member 2025-present; Vice President 2026)

- Honors Undergraduate Social Committee at University of New Hampshire (2023-present; Secretary 2025-present)

- Surfrider Foundation Chapter at University of New Hampshire (2025-present)

- Intramural Soccer at University of New Hampshire (2023-present; Team Captain 2024; 2025)

- Fencing Club at University of New Hampshire (2023-present)

- Perry the Platypus Club at University of New Hampshire (2024)

- Zoological Society member at University of New Hampshire (2023-2025)

- Intramural Volleyball at University of New Hampshire (2023)

- Leadership Academy member (2023)

- Engineering Club Secretary (2022-2023)

- Varsity Cross Country Leader (2021-2023)

- Varsity Soccer Captain (2020-2023)

- Extreme Math Team Captain (2020-2023)

- Quiz Bowl Varsity Team Captain (2019-2023)

OUTREACH

Exeter Alewife Festival Exeter, NH **May 9, 2025**

- I tabled with Dr. Nathan Furey's Fish and Movement Ecology Lab chatting with hundreds of members of the local community about river herring and the research we do in the Gulf of Maine and nearby river systems. We specifically focused on tracking technology, how to tell alewife from blueback herring non-lethally (their scales), and the role of river herring in local ecosystems as food for terns and striped bass, other species we study.

Ocean Discovery Day at University of New Hampshire Durham, NH **Annual October event: 2025-2026**

- Worked with Dr. Nathan Furey's Fish and Movement Ecology Lab to collectively educate thousands of students, families, and college students on Acoustic Telemetry, fish tagging, elasmobranch teeth and jaws, and seabird movement ecology and tracking. We connected with a diverse group of people to excite them about oceans and ocean life